

## Project Initialization and Planning Phase

|               |             |
|---------------|-------------|
| Date          | 31 May 2026 |
| Team ID       | xxxxxxx     |
| Project Name  | Smart Crop  |
| Maximum Marks | 3 Marks     |

### Define Problem Statements (Customer Problem Statement Template):

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love. A well-articulated customer problem statement allows you and your team to find the ideal solution for your customers' challenges. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

### Example:

| Problem Statement (PS) | I am (Customer)                                                    | I'm trying to                                                                             | But                                                                       | Because                                                                      | Which makes me feel                                                |
|------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------|
| PS-1 (Scenario 1)      | A local traditional farmer with varying soil health profiles.      | Maximize crop yields and resource efficiency during this planting season.                 | I lack data-driven insights into specific dynamic variables.              | Nitrogen (N), Phosphorous (P), Potassium (K), and pH levels change wildly.   | Anxious and uncertain about crop selection and farm profitability. |
| PS-2 (Scenario 2)      | An enterprise agricultural grower evaluating crop diversification. | Assess environmental suitability and productivity potential for a new targeted crop type. | Manually predicting alignment across erratic microclimates is inaccurate. | Temperature, humidity, and rainfall factors interact in highly complex ways. | Frustrated by high financial risk and potential resource waste.    |

|                      |                                                              |                                                                                             |                                                                            |                                                                           |                                                                            |
|----------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------|
| PS-3<br>(Scenario 3) | An agricultural researcher or government policy stakeholder. | Establish evidence-based sustainable farming strategies and resource optimization programs. | Macro patterns in regional crop-environment interactions remain obscured.. | Legacy data tools lack integrated predictive engines or visual frameworks | Disconnected from real farming realities when formulating master policies. |
|----------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------|